Department of Education

Computer System Servicing (CSS12)

INSTALLING AND CONFIGURING COMPUTER SYSTEMS (ICCS)

Module 1

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What I Need to Know

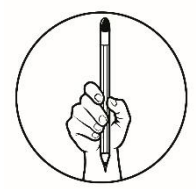
This module was designed and written with you in mind. It is here to help you master the nature of Computer System Servicing. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course.

The module is divided into four lessons, namely:

- Lesson 1 - Assemble Computer Hardware
- Lesson 2 - Prepare installer
- Lesson 3 - Install operating system and drivers for peripherals/devices
- Lesson 4 - Install application software

After going through this module, you are expected to:

1. identify the parts and functions of computer;
2. describe and perform computer assembly procedures;
3. describe and differentiate the different types of Operating System;
4. prepare and install operating system, hardware drivers and softwares.



What I Know

Pre-Assessment 1.1

Choose and Encircle the letter of the best answer.

1. It is the point where power enters your system from an external power source
   a. Hard Disk
   b. Power Supply Unit
   c. Central Processing Unit
   d. Graphic Processing Unit

2. Responsible for processing all information from programs run by your computer
   a. Hard Disk
   b. Power Supply Unit
   c. Central Processing Unit
   d. Graphic Processing Unit

3. Especially important for 3D rendering, the GPU does exactly what its name suggests and processes huge batches of graphic data
    a. Hard Disk
    b. Power Supply Unit
    c. Central Processing Unit
    d. Graphic Processing Unit

4. The main circuit board of your computer and is also known as the mainboard or logic board
    a. Motherboard
    b. Heat Sink/Fan
    c. Back Panel
    d. BIOS

5. Is a storage device responsible for storing permanent and temporary data
    a. Random Access Memory
    b. Central Processing Unit
    c. Hard Drive
    d. Power Supply Unit

6. Temporarily store on-the-fly information
    a. Random Access Memory
    b. Central Processing Unit
    c. Hard Drive
    d. Power Supply Unit

7. Physical components that a computer system requires to function
    a. Keyboard
    b. Mouse
    c. Motherboard
    d. Computer Hardware

8. Drives that are installed in internal bays are called
    a. Computer Hardware
    b. Internal Drives
    c. External Drives
    d. Hard Drive

9. Example of internal drive:
    a. Hard Drive
    b. Motherboard
    c. Mouse
    d. Keyboard

10. Transmit data between the motherboard and storage devices
    a. Power Supply Unit
    b. Central Processing Unit
    c. Power Cable
    d. Data Cable

Lesson 1 Assemble Computer Hardware

Computer Hardware - in the computer world, refers to the physical components that make up a computer system.

Although the design of hardware differs between desktop PCs and laptops due to their differences in size, the same core components will be found in both. Without hardware, there would be no way of running the essential software that makes computers so useful. Software is defined as the virtual programs that run on your computer; that is, operating system, internet browser, word-processing documents, etc.

Although a computer can function only when both hardware and software are working together, the speed of a system will largely rely on the hardware used.

When building up a new computer, or simply replacing old parts, you may need to know the specific hardware in your computer. The purpose of this guide is therefore to help you understand the inner-workings of your computer.

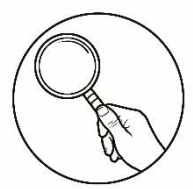
What's In

On the previous lesson you learn on how to obtain and clarify work instructions based on job order or client requirements. In this lesson you will learn how to assemble computer hardware based on the client requirements.

What's New

Computer Hardware

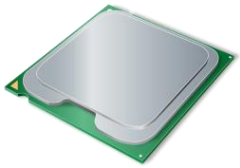
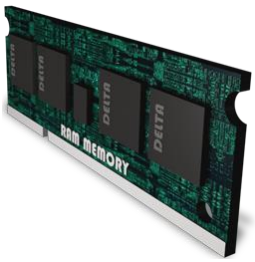
Quite simply, computer hardware is the physical components that a computer system requires to function. It encompasses everything with a circuit board that operates within a PC or laptop; including the motherboard, graphics card, CPU (Central Processing Unit), ventilation fans, webcam, power supply, and so on.

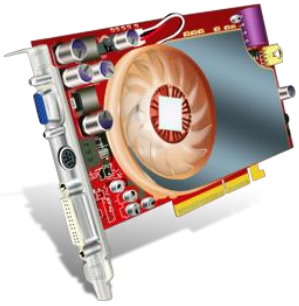
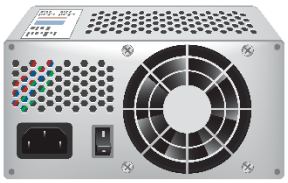


What is It

Common Individual Hardware Components

Table 1. Some Individual Hardware Components

Hardware Name	***Function***
Motherboard	the main circuit board of your computer and is also known as the mainboard or logic board
CPU (Central Processing Unit or processor) 	responsible for processing all information from programs run by your computer
Random Access Memory (RAM) 	The role of RAM is to temporarily store on-the-fly information created by programs and to do so in a way that makes this data immediately accessible

Hard Drive	is a storage device responsible for storing permanent and temporary data
Graphics Processing Unit (GPU) 	Especially important for 3D rendering, the GPU does exactly what its name suggests and processes huge batches of graphic data
Power Supply Unit (PSU) 	It is the point where power enters your system from an external power source and is then allocated by the motherboard to individual component hardware

Computer assembly: Step-by-Step

Caution: Unplug power supply before performing the procedures. For the safety of the computer make sure to discharge static electricity or use anti-static wrist wrap.

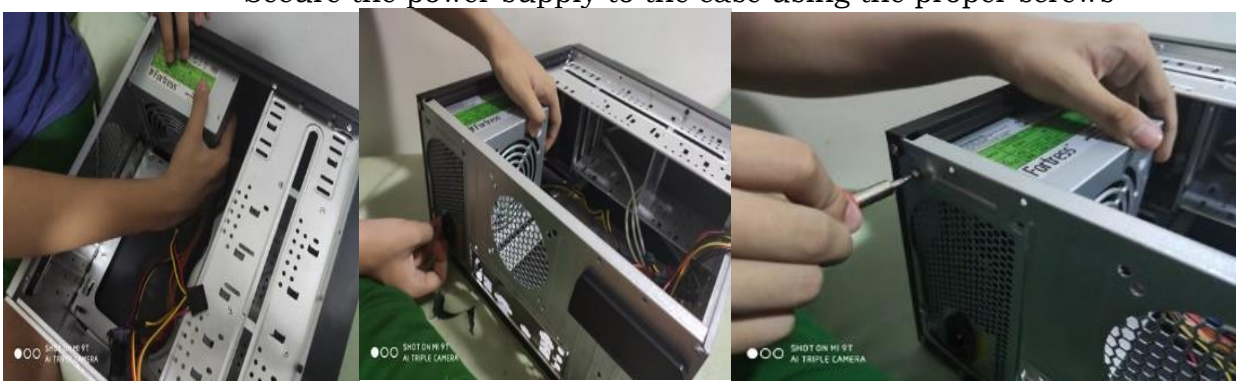
I. Open the Case

i. Computer assembly is a large part of a technician's job.
ii. Prepare the workspace before opening the computer case.
iii. There are different methods for opening cases. To learn how, consult the user manual or manufacturer's website.

II. Install the Power Supply

i. Power supply installation steps include the following:

- Insert the power supply into the case
- Align the holes in the power supply with the holes in the case
- Secure the power supply to the case using the proper screws

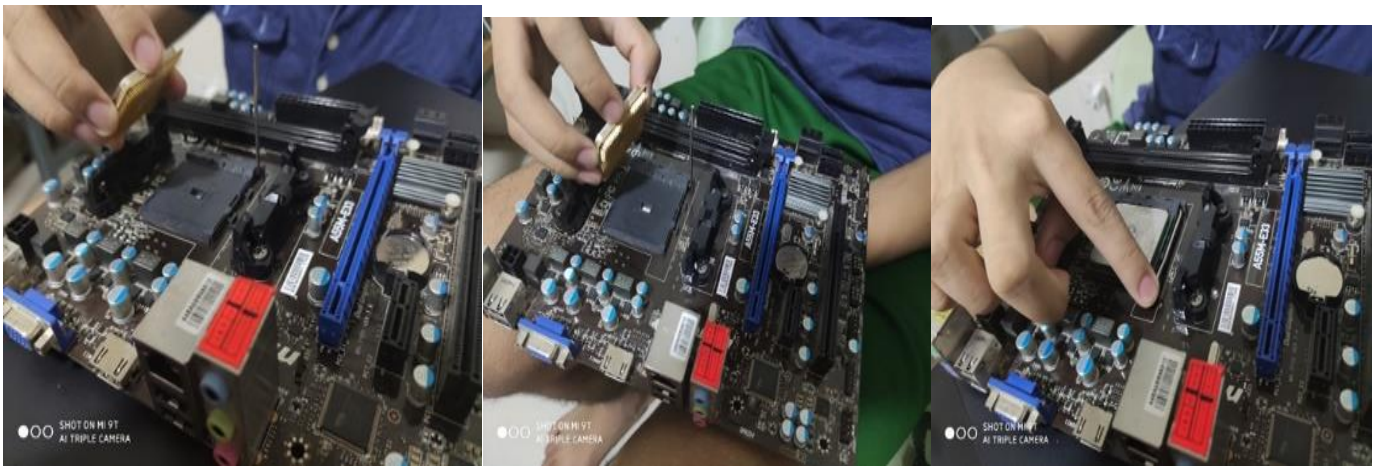


III. Attach Components to the Motherboard

i. CPU on Motherboard

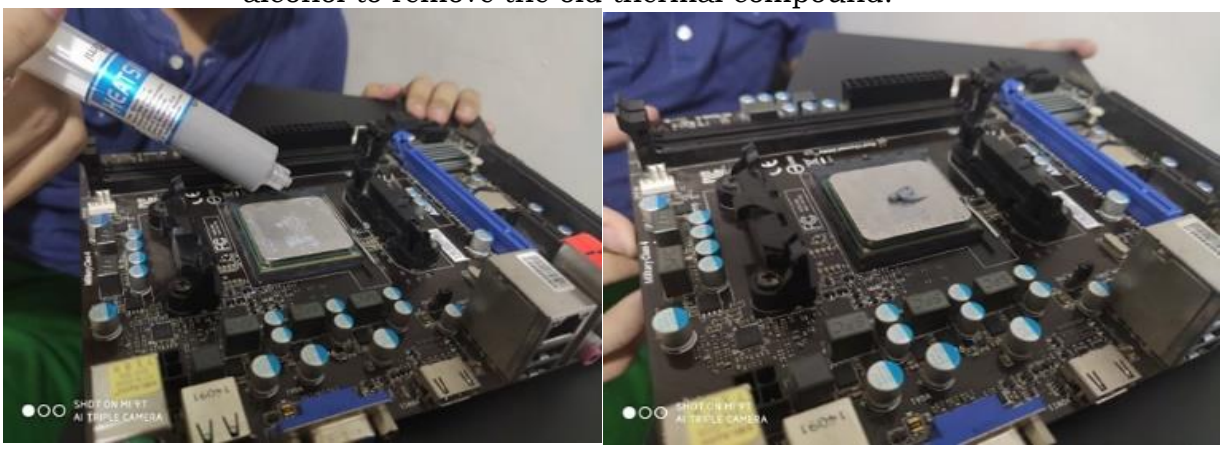
- The CPU and motherboard are sensitive to electrostatic discharge.
- The CPU is secured to the socket on the motherboard with a locking assembly.

CAUTION: When handling a CPU, do not touch the CPU contacts.



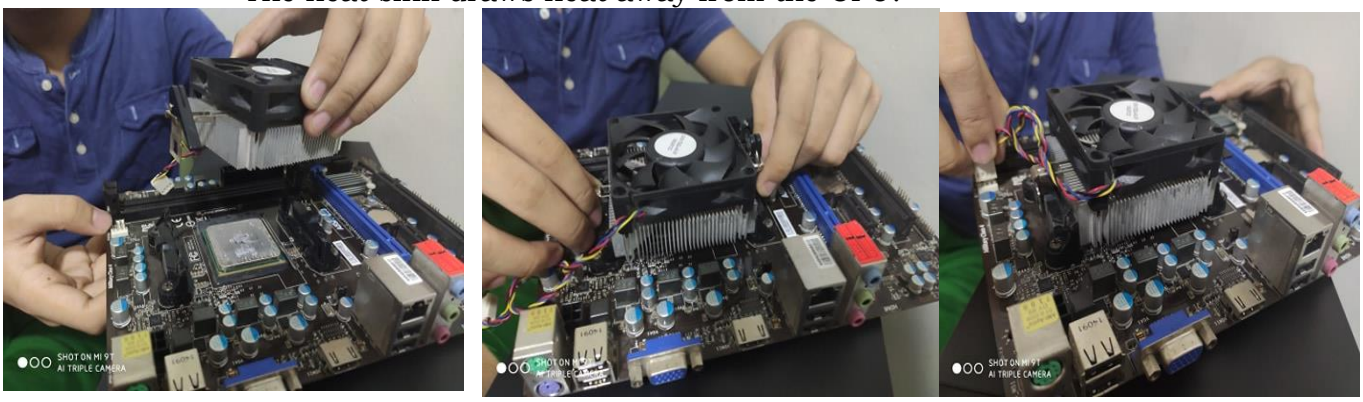
ii. Thermal compound

- It helps to keep the CPU cool.
- To install a used CPU, clean it and the base of the heat sink with isopropyl alcohol to remove the old thermal compound.



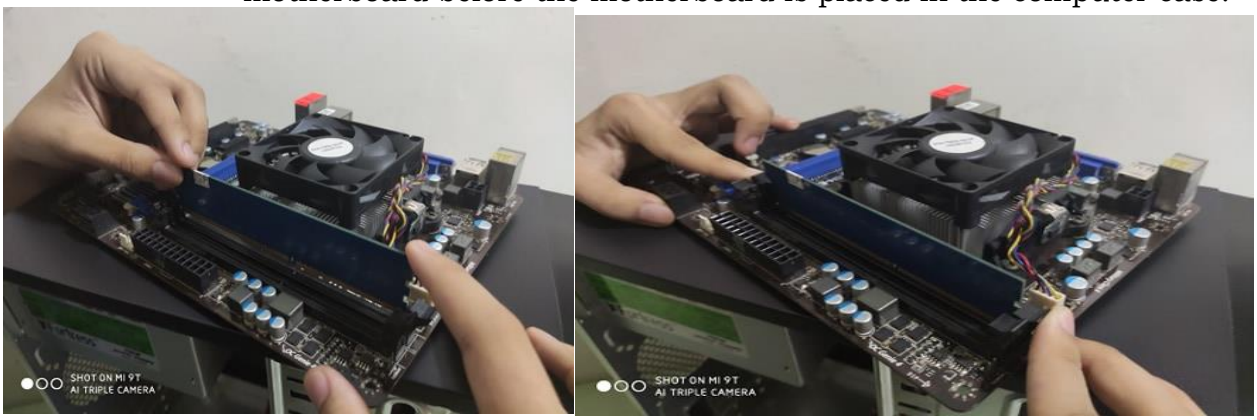
iii. Heat Sink/Fan Assembly

- The Heat Sink/Fan Assembly is a two-part cooling device.
- The heat sink draws heat away from the CPU.



iv. Install RAM

- RAM provides temporary data storage for the CPU and should be installed in the motherboard before the motherboard is placed in the computer case.



IV. Install the Motherboard

i. After installing the previous components the motherboard is now ready to install in the computer case.

ii. Plastic and metal standoffs are used to mount the motherboard and to prevent it from touching the metal portions of the case.

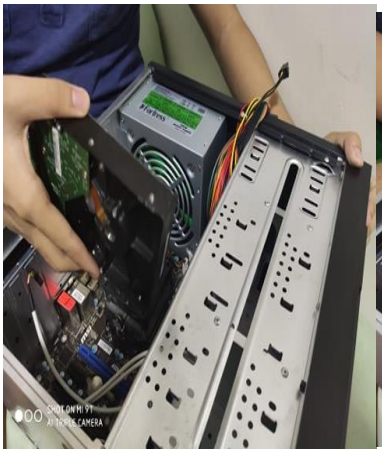
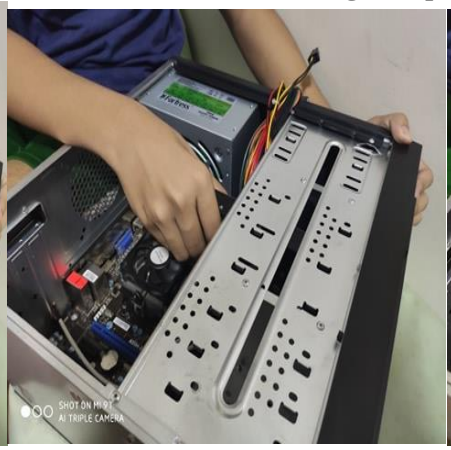
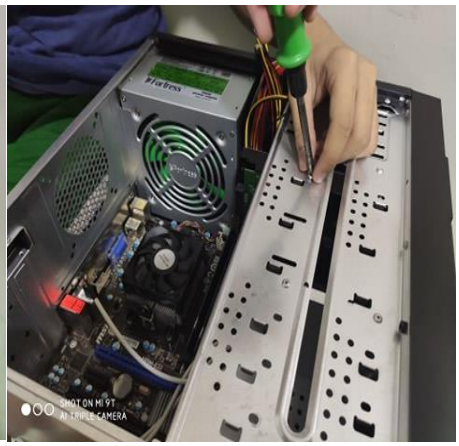
V. Install Internal Drives

i. Drives that are installed in internal bays are called internal drives.

ii. A hard disk drive (HDD) is an example of an internal drive.

iii. HDD installation steps:

- Position the HDD so that it aligns with the 3.5-inch drive bay.
- Insert the HDD into the drive bay so that the screw holes in the drive line up with the screw holes in the case.
- Secure the HDD to the case using the proper screws.

VI. Install Drives in External Bays

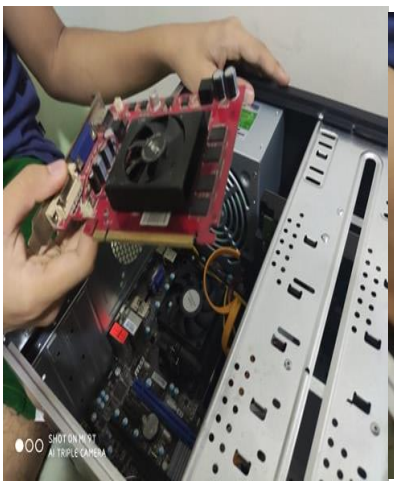
i. Drives in external bays allow access to the media without opening the case.

ii. Some devices that are installed in this type of drives are:
- An optical drive is a storage device that reads and writes information to CDs or DVDs.
- A floppy disk drive (FDD) is a storage device that reads and writes information to a floppy disk.

VII. Install Adapter Cards

i. Adapter cards are installed to add functionality to a computer.

ii. Some examples of these adapters are:
- A NIC which enables a computer to connect to a network. It uses PCI and PCIe expansion slots on the motherboard.
- A wireless NIC which enables a computer to connect to a wireless network. Wireless NICs use PCI and PCIe expansion slots on the motherboard. Some wireless NICs are installed externally with a USB connector.
- A video adapter card is the interface between a computer and a display monitor. An upgraded video adapter card can provide better graphic capabilities for games and graphic programs. Video adapter cards use PCI, AGP, and PCIe expansion slots on the motherboard.



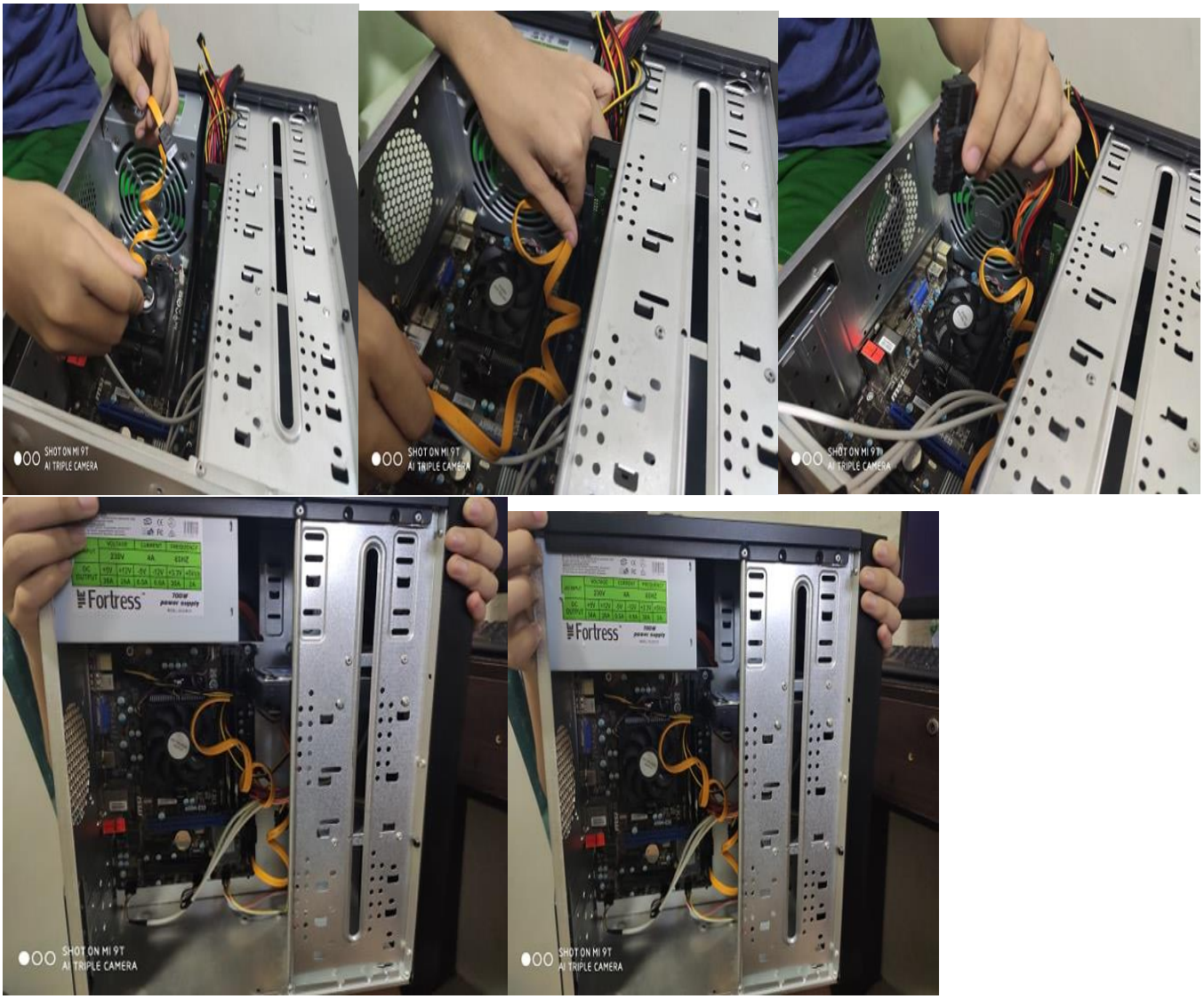
VIII. Connect Internal Cables

i. Power cables are used to distribute electricity from the power supply to the motherboard and other components.

ii. Data cables transmit data between the motherboard and storage devices, such as hard drives. Some of examples of this type of cables are:

- PATA cable
- SATA cable
- Floppy drive data cable

iii. Additional cables connect the buttons and link lights on the front of the computer case to the motherboard.



IX. Complete Physical Installation

i. Now that all the internal components and the power supply have been installed and connected to the motherboard, the following tasks should be completed:

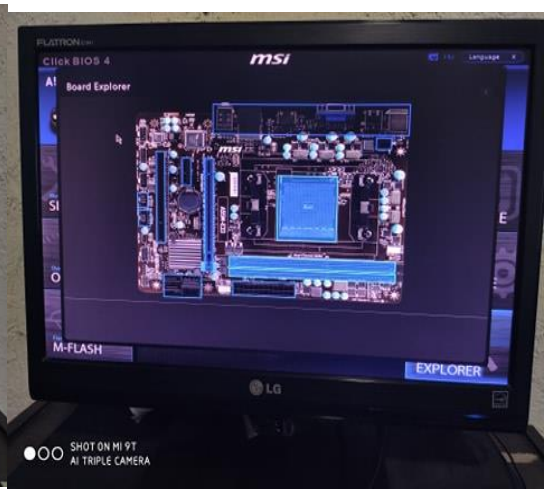
- Re-Attach the side panels: Most computer cases have two panels, one on each side. Some computer cases have one three-sided cover that slides down over the case frame.
- Connect External Cables: These cables are normally connected to the back of the computer. Here are some common external cable connections: Monitor, Keyboard, Mouse, USB, Ethernet Power.

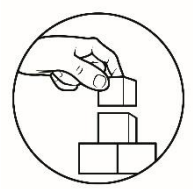
X. Boot Computer for the First Time

i. When the computer is booted, the basic input/output system (BIOS) will perform a power-on self test (POST) to check on all of the internal components.

ii. The BIOS contains a setup program used to configure settings for hardware devices. The configuration data is saved to a special memory chip called a complementary metal-oxide semiconductor (CMOS).

iii. POST checks to see that all of the hardware in the computer is operating correctly. If a device is malfunctioning, an error or a beep code alerts the technician that there is a problem.





What's More

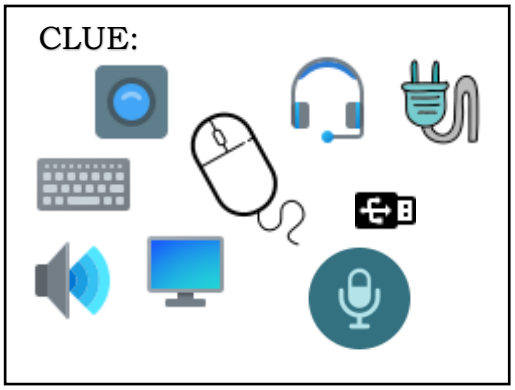
Activity 1.1 Understanding Computer Hardware

Can you solve the hidden word?

Unscramble the names of COMPUTER HARDWARE to find the hidden word.

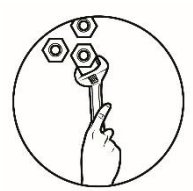
1. ROOINMT
2. AEKSPSRE
3. RMEAAWCEB
4. ETSHADE
5. PCEROMIOHN
6. USEMO
7. SHRVILDAFE
8. DWEPCOROR
9. RDKOEAYB

CLUE:



What I Have Learned

What do you think are the benefits of learning the process and procedures of assembling computer hardware? On what way do you think it will help you in the future? And what is the most challenging part of the process of assembling computer? Write your answers on a sheet of paper.



What I Can Do

Create a collage or video performing the assembling of computer hardware with proper precautions and following the step-by-step procedures given.

Assessment

Assessment 1.1

Answer each questions with True or False.

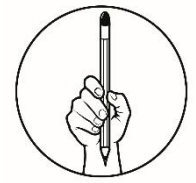
1. Computer Hardware - in the computer world, refers to the physical components that make up a computer system.
2. Power cables are used to distribute data.
3. Adapter cards are installed to add functionality to a computer.
4. BIOS means Basic Input/Output System.
5. CMOS means complementary metal-oxide semiconductor.
6. Hard Drive provides temporary data storage for the CPU.
7. RAM draws heat away from the CPU.
8. Thermal Compound helps to keep the CPU cool.
9. Computer assembly is a large part of a technician's job.
10. The CPU and motherboard are sensitive to electrostatic discharge.

Additional Activities

- Watch this video for reference:
  https://www.facebook.com/jdmf12/videos/vb.1157073359/10220087613878088/?type=3

- Set-up a timer and perform the procedures of assembling computer hardware and record your time from the start up to the last part of the procedures. Try to beat your fastest time and set a new record.

Lesson 2 Preparation on Installing Operating System and other devices



What I Know

Pre-Assessment 1.2

Answer each questions with True or False.

1. An operating system is a set of programs that manage computer hardware resources and provide common services for the application software.
2. Operating system is not a vital component of system software in a computer system. Application programs require an operating system which are usually separated programs, but can be combined in simple systems.
3. Time-sharing operating system schedule time for efficient use of the system and may also include accounting for cost allocation of processor time, mass storage, printing, and other resources.

4. For software functions such as input and output and memory allocation, the operating system act as an intermediary between application programs and the computer hardware.
5. Operating systems are found on almost any device that contains a computer from cellular phones and video games consoles to supercomputers and web servers.

What's In

On the previous lesson you learn to identify the parts of computer and how to assemble computer hardware. After the assembling of computer hardware we will now go to preparing the installation of operating system (OS) and other devices.

What's New

Before we move on, let us first understand the following terms which will help us in understanding the lesson

Definition of terms

Boot is to load the first piece of software that starts a computer. Because the operating system is essential for running all other programs, it is usually the first piece of software loaded during the boot process.

Storage Media is a device for recording (storing) information (data).

Bootable Media/ Bootable Device is physical media (CD, DVD, USB flash drive or other removable media supported by a machine BIOS as a boot device) that boots on any PC-compatible machine.

Bootable media is most often used to:
• recover an operating system that cannot start
• access and back up the data that has survived in a corrupted system
• back up sector-by-sector a disk with an unsupported file system
• hardware or software troubleshooting

Warm Boot vs. Cold Boot

• Cold Boot
To perform a cold boot (also called a "hard boot") means to start up a computer that is turned off. It is typically performed by pressing the power button on the computer.

• **Warm Boot**
To perform warm boot (also called a "soft boot") means to restart a computer. It is typically initiated from the operating system, rather than pressing a button on the computer itself.

For example, to perform a warm boot on a Windows system, you may select Shut Down → Restart from the Start Menu. If you use a Mac, you can perform a warm boot by selecting Restart… from the Apple Menu

What is It

Operating system is a system software commonly simplified as “OS”. An operating system is the most important software that runs on a computer. It manages the computer's memory, processes, and all its software and hardware. It also allows you to communicate with the computer without knowing how to speak the computer's "language." Without an operating system, a computer is useless.

The Operating System's Job
Computer's operating system (OS) manages all of the software and hardware on the computer. Most of the time, there are several different computer programs running at the same time, and they all need to access your computer's central processing unit (CPU), memory, and storage. The operating system coordinates all of this to make sure each program gets what it needs.

TYPES OF OPERATING SYSTEMS
Operating systems usually come preloaded on any computer that you buy. Most people use the operating system that comes with their computer, but it is possible to upgrade or even change operating systems.

The three most common operating systems for personal computers are Microsoft **Windows**, **Apple Mac OS X**, and **Linux**.

Microsoft Windows
Microsoft created the Windows operating system in the mid-1980s. Over the years, there have been many different versions of Windows, but the most recent ones are Windows 10 (released in 2015), Windows 8 (2012), Windows 7 (2009), and Windows Vista (2007). Windows comes pre-loaded on most new PCs, which helps to make it the most popular operating system in the world.

Apple Mac OS X
Mac OS is a line of operating systems created by Apple. It comes preloaded on all new Macintosh computers, or Macs. All of the recent versions are known as OS X (pronounced O-S Ten), and the

specific versions include El Capitan (released in 2015), Yosemite (2014), Mavericks (2013), Mountain Lion (2012), and Lion (2011).

Linux
It is a family of open source operating systems, which means that they can be modified and distributed by anyone around the world. This is very different from proprietary software like Windows, which can only be modified by the company that owns it (Microsoft). The advantages of Linux are that it is free, and there are many different distributions (or versions) that you can choose from. Each distribution has a different look and feel, and the most popular ones include Ubuntu, Mint, and Fedora.

Linux is named after Linus Torvalds, who created the Linux kernel in 1991. The kernel is the computer code that is the central part of an operating system.

Modern operating systems use a **Graphical User Interface**, or GUI **(pronounced "gooey")**. A GUI lets you use your mouse to click on icons, buttons, and menus, and everything is clearly displayed on the screen using a combination of graphics and text.

OPERATING SYSTEMS FOR MOBILE DEVICES
The operating systems that we've been talking about were designed to run on desktop or laptop computers. Mobile devices such as phones, tablet computers, and mp3 players are very different from desktop and laptop computers, so they run operating systems that are designed specifically for mobile devices. Examples of mobile operating systems include **Apple iOS, Windows Phone, and Google Android.**

What is It

CREATE WINDOWS INSTALLER USING FLASH DRIVE
Small portable computing devices such as Netbooks and MIDs (Mobile Internet Devices) offer great mobility, but they lack optical drives (CD/DVD). An optical drive is important when you need to install software from a CD or DVD. An external optical drive would work, but you may not want to lug it around with you. So what's the solution? Use a USB flash drive as a bootable installation media.

What You Need:

1. A USB flash drive with good storage capacity; preferably 4GB or more.
2. Access to a computer running Windows Vista or Windows 7 or Windows 10.
3. A CD or DVD or IMG file containing software you want to download.

Procedure

1. Click Start > All Programs > Accessories > Right click on Command Prompt > Run as Administrator.

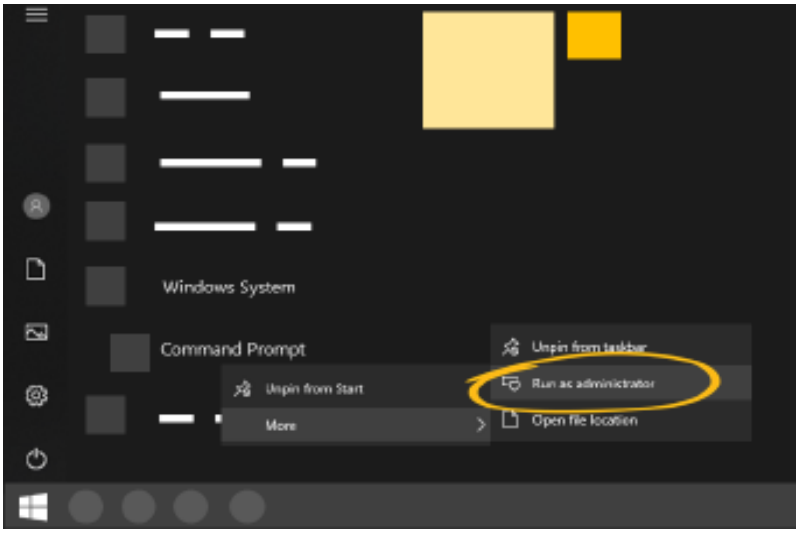


Figure 1.0 ***Running CMD with Administrative privilege***

2. Type 'diskpart' and press Enter to start the built in Windows disk partitioning utility

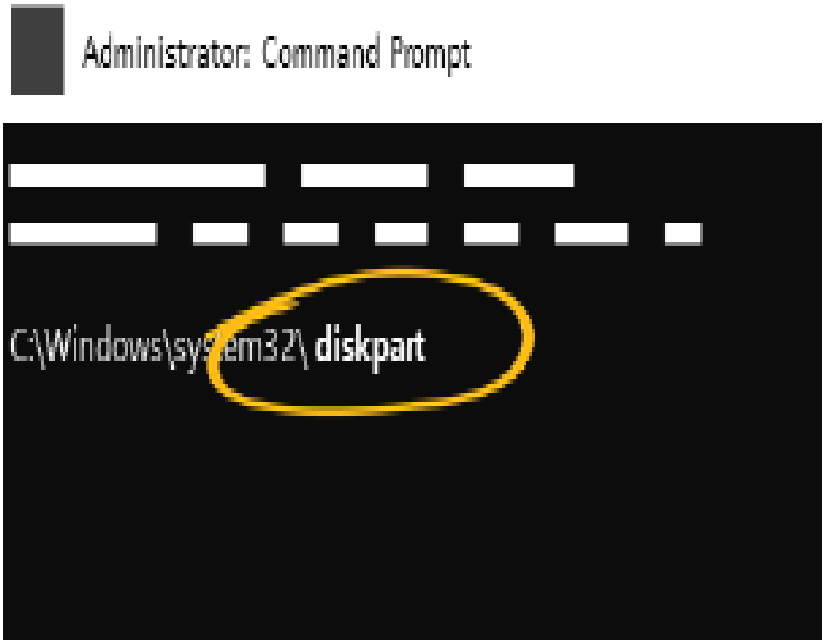


Figure 1.1 ***Starting diskpart portioning utility***

3. Type 'list disk' and press Enter. Identify your USB Flash drive and make a note of the disk number. Here in this example, My USB flash drive is Disk 4.

Note: Careful about choosing correct disk index number, failure to select right disk may result to permanent deletion of data that may lead to catastrophic event.

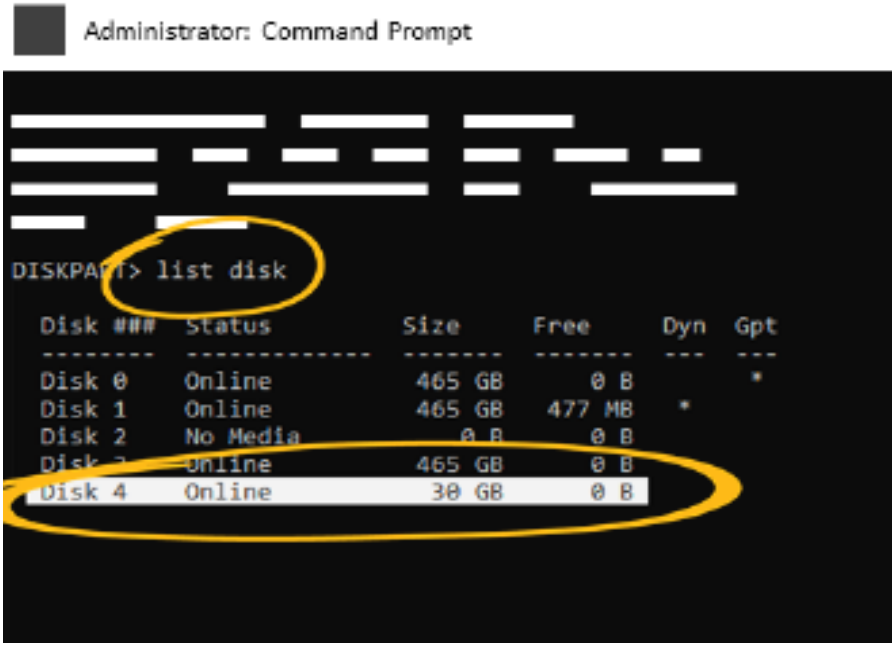


Figure 1.3 ***Listing available disk***

4. Type 'select disk 4' and press Enter.

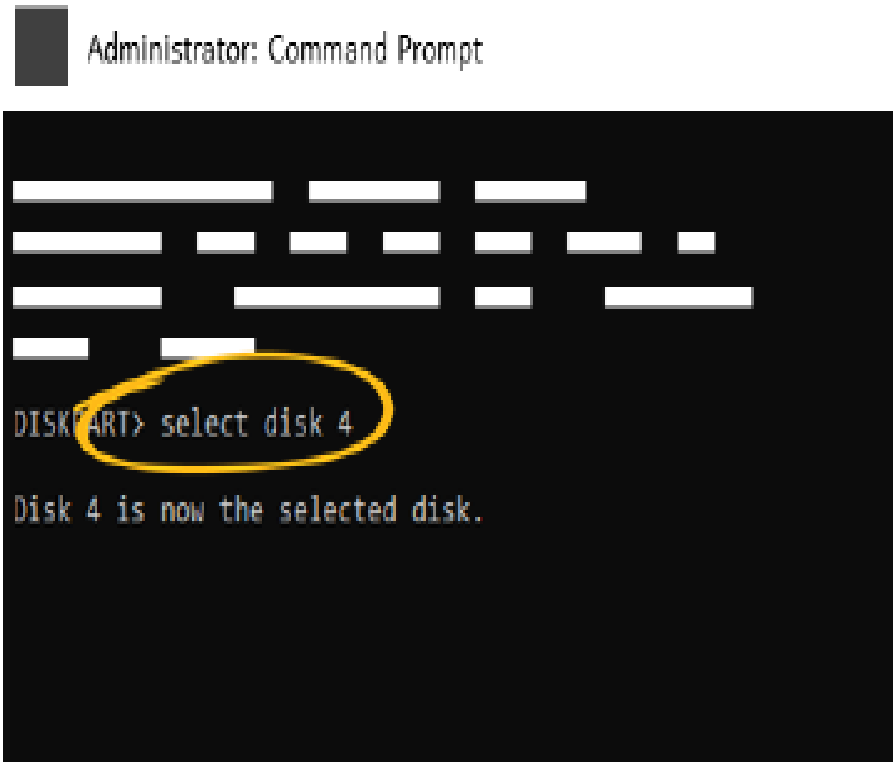


Figure 1.4 ***Running CMD with Administrative privilege***

5. Type ‘clean’ and press Enter. The clean command will remove partitions or volume formatting from the selected disk (disk 1) by zeroing sectors.

Figure 1.5 ***clean Command***

6. Now to create a partition, type 'create partition primary' and press Enter.

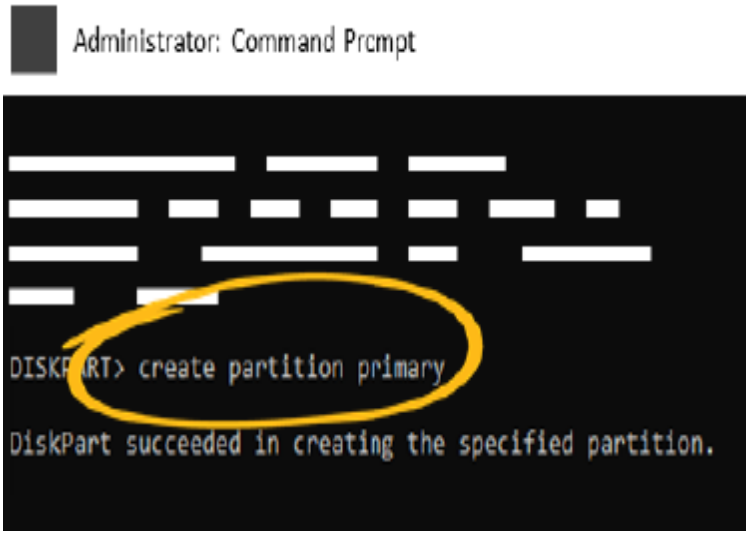


Figure 1.6 ***Create primary partition***

7. Type 'select partition 1', and press the Enter key

Figure 1.7 ***Select Partition***

8. Type ‘active’, and hit the Enter Key.

Figure 1.8 ***active selected partition***
Source: http://www.support.com/

9. To format the partition using FAT32 file system, type ‘format fs=fat32’ and press the Enter key.

Important
If your server platform supports Unified Extensible Firmware Interface (UEFI), you should format the USB flash drive as FAT32 rather than as NTFS. To format the partition as FAT32, type format fs=fat32 quick, and then click ENTER.

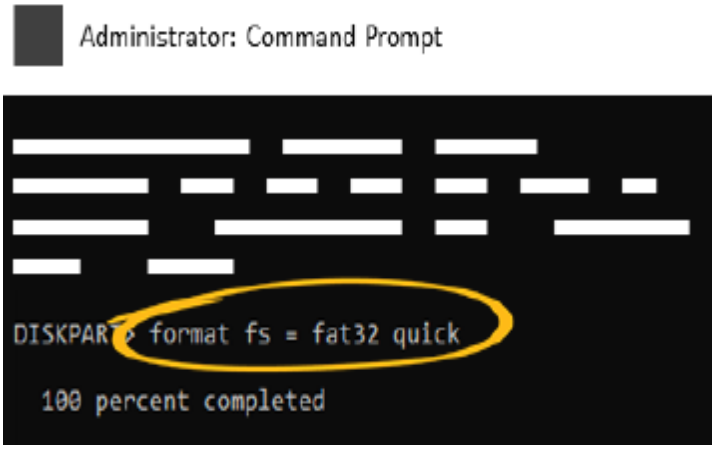


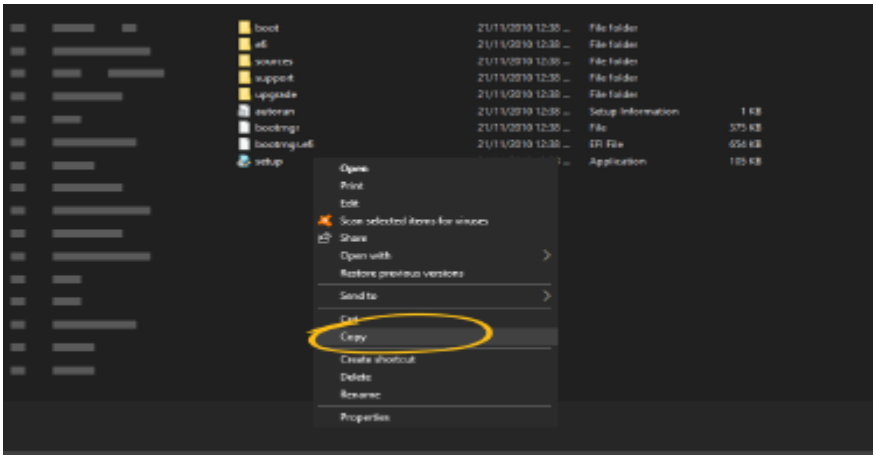
Figure 1.9 ***format FAT32 file system***

10. Type 'assign' and press the Enter key to assign a drive letter to your USB flash drive. Type exit to exit the diskpart utility.

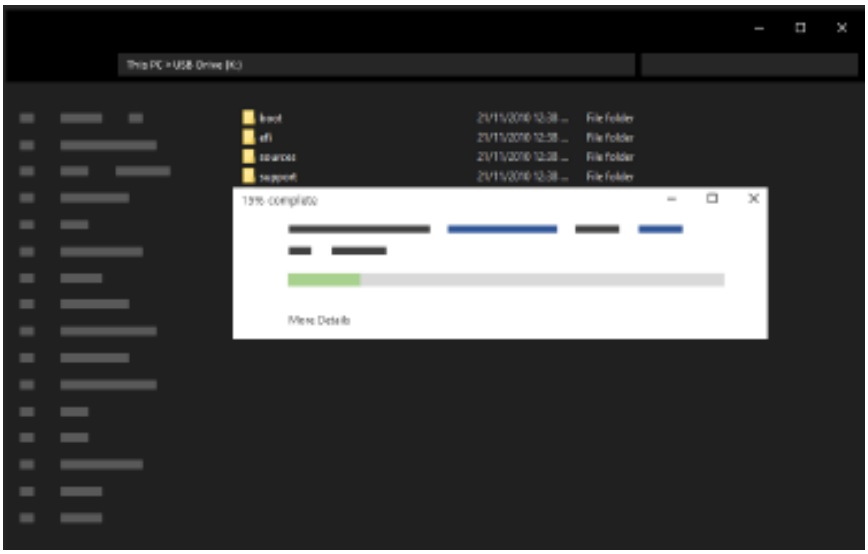
Figure 1.10 ***Assign driver letter***

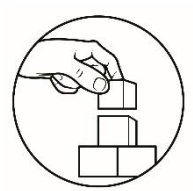
Your USB flash drive is now ready to be used as an installation media. All you need to do is copy the entire contents of the Operating System (DVD) to your newly prepared USB flash drive. Plug the USB flash drive to the computer and reboot to begin the installation.

11. Copy the data.



12. Paste the data file into your flashdrive.





What's More

Activity 1.2 Choose and Encircle the letter of the best answer.

1. **It is a device for recording (storing) information (data).**
   a. Storage Media
   b. Operating system
   c. Boot
   d. Application

2. **It is also called a "hard boot".**
   a. Cold boot
   b. Warm Boot
   c. Storage Media
   d. Flash drive

3. **It is also called a "soft boot".**
   a. Cold boot
   b. Warm Boot
   c. Storage Media
   d. Flash drive

4. **Bootable media is most often used to.**
   a. recover an operating system that cannot start
   b. back up sector-by-sector a disk with an unsupported file system
   c. both A and B
   d. install antivirus

5. **It pertains to any physical media (CD, DVD, USB flash drive or other removable media supported by a machine BIOS as a boot device) that boots on any PC-compatible.**
   a. Bootable devices
   b. Operating system
   c. both A and B
   d. application software

6. **Means to start up a computer that is turned off. It is typically performed by pressing the power button on the computer.**
   a. Cold Boot
   b. Warm Boot
   c. bootstraps
   d. bootable disc.

7. **Means to restart a computer. It is typically initiated from the operating system, rather than pressing a button on the computer itself.**
   a. bootstraps
   b. Cold Boot
   c. bootable disc
   d. Warm Boot

8. **Is to load the first piece of software that starts a computer.**
   a. application software
   b. bootstraps
   c. boot
   d. Flash drive

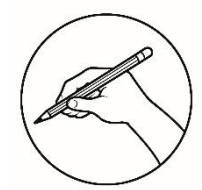
9. **Is physical media (CD, DVD, USB flash drive or other removable media supported by a machine BIOS as a boot device) that boots on any PC-compatible machine.**
   a. Flash drive
   b. Bootable Media

c. Storage Media

10. It pertains to load the first piece of software that starts a computer
a. Boot
b. Operating system
c. Storage Media
d. bootstraps

What I Have Learned

In this lesson you learned that there are 3 types of Operating Systems. What do you think is the best to use? Why? Write your answer on a separate sheet of paper.



Assessment

Assessment 1.2

I. Multiple Choice Directions: Read each question and their corresponding answers carefully and completely. Choose the answer that best fits the question. Please circle the appropriate letter.

1. What is OS?
a. Operating System
b. Operation System
c. Operate System
d. Operation Procedure

2. What is the correct sequence of diagram describes the operation of operating system.
a. Hardware-Apps-User-OS
b. OS-User-Apps-Hardware
c. User-Apps-OS-Hardware
d. OS-User-Hardware-Apps

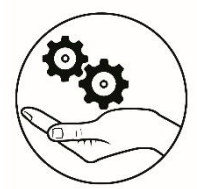
3. It lets you use your mouse to click on icons, buttons, and menus, and everything is clearly displayed on the screen using a combination of graphics and text.
a. CLI
b. GUI
c. OS
d. Apps

4. It manages the computer's memory, processes, and all of its software and hardware.
a. Operating system

b. software
c. Windows 7
d. Application Software

5. According to StatCounter Global Stats, Linux users account for less than __% of the operating systems market as of January 2013.
a. 1%
b. 10%
c. 100%
d. 50%

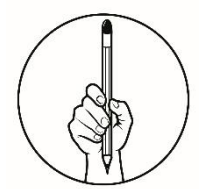
II. Enumeration: Enumerate the following.
1-3. Give three mobile operating system
4-5. Give two desktop operating system.



Additional Activities

- Watch this video for reference:
  https://www.youtube.com/watch?v=OJ5Teoiicy8
  https://www.facebook.com/cssnc2/videos/2121825097864065/
- Create a bootable device for windows 7 and windows 10. Record the process of creating a bootable media.

Lesson 3 Install Operating System and Drivers for peripherals/ devices



What I Know

Pre-Assessment 1.3

Answer each questions with True or False.

1. Windows is a type of operating system.
2. You can't create a bootable media on CD, DVD and other storage device except for USB flash drive.
3. Partitioning is part of the process of installing operating system.
4. You can install 2 operating systems in one computer.
5. If one of the system's drivers was not installed the hardware may not work properly according to it's function.

What's In

On the previous lesson you learn how to prepare and create bootable media using USB flash drive and you also discover the function and use of the Operating System (OS). In this lesson, you will learn how to execute the bootable device and how to install operating system and other driver installers.

What's New

A device is disabled without a driver.

Compare to real life scenario a device will not perform its specific task without a device driver or a software that acts as an activator to activate certain device in order to perform its full functionality.

In this information, you are about to learn the definition of driver, its function and its purpose in conducting software installation. How to prepare driver for installation.

DRIVER

A driver or device driver is a software program that enables a specific hardware device to work with a computer's operating system.

Figure 3.1 ***Driver's function***

Drivers may be required for internal components, such as video cards and optical media drives, as well as external peripherals, such as printers and monitors.

But most modern hardware is "plug and play" meaning the devices will work without requiring driver installation. However, even if a hardware device is recognized by the operating system, installing the correct drivers may provide additional options and functionality for the device.

For example, most mice work automatically when they are connected to a PC. However, installing the appropriate mouse driver may allow you to customize the function of each button and adjust the mouse sensitivity. Some keyboard drivers allow you to assign functions to specific keys, such as controlling the volume or opening specific applications.

Driver is the linker of a certain hardware to computer. A hardware without driver is like a plane without a pilot. It is still a machine but it does not act like a machine itself.

Source: http://www.techterms.com

Before installation/reformatting we should check first all device drivers needed in installation process. Failure to do so, it may leads to different errors that may cause the computer and its connected devices to work improperly. The following errors may occurs if device drivers are not properly installed.

1. If system's drivers are not installed.
a. No sounds input and output (mic and speaker are not working properly)
b. Not able to connect to the network
c. Poor video display
d. Manufacturers default software are not possible to use
e. Not able to print, if printer's driver is not installed
f. And some IO ports are not usable (e.g, sd card reader)

2. If add-on devices' drivers are not installed, including input and output devices.

a. Devices are not possible to work properly according to its specificity.

E.g., Scanner is not able to scan without the presence of its required driver.

Preparing Driver for Installation
Procedures in preparing driver for installation.

1. Check the installation driver disc attached in a newly purchased computer.
2. If don't have driver disc, try to identify the specificity of a computer, including its brand name

and code then search over the internet the equivalent needed drivers.
3. If internet connection is not available, make it sure that you prepared the all in one (AIO) DriverPack Solution app for you to install the required drivers for installation.

Choosing the right drivers

In choosing the correct drivers, you must to consider the following:

Check the specific required drivers for its equivalent devices.
E.g. You lose your disc driver attached upon purchasing your laptop.

You are trying to reformat a laptop. The first thing you need to do is to obtain its hardware specification, get the specific brand name and code of your device. Search the code in the internet and find its exact specification and download the drivers.

Every devices has an equivalent drivers, except those Plug and Play (PnP) devices which do not requires drivers in order to use it.

What is It

Why Boot a Windows Installation From USB?

If your spare PC doesn't have an optical drive, or you're out of DVDs, a bootable USB stick is the best solution.

After all, a USB stick is portable, and you can guarantee it is compatible with every desktop and laptop computer. While some computers might be missing a DVD drive, they all have a USB port.

It's also quicker to install from USB. A USB drive can be made bootable faster than an optical drive can; it also installs the operating system quicker.

Ensuring USB Stick Has UEFI Boot Support

Before you download a bootable Windows installation image, it's important to know the difference between UEFI and BIOS.

Older PCs rely on the basic input/output system (BIOS) to boot the operating system and manage data between the operating system and devices. For the past decade, UEFI (Unified Extensible Firmware Interface) has replaced BIOS, adding legacy support. UEFI can assist with PC diagnosis and repair without additional software or media.

Fortunately, the most common methods of installing Windows 10 from a bootable USB drive support UEFI and legacy BIOS hardware. So, whichever option you choose should work for your hardware.

Installing Windows 10 with a Bootable USB Drive (STEP-BY-STEP)

With the installation media created, you'll be able to install Windows 10. The USB drive is now bootable, so all you need to do is safely remove it from your PC, then insert it into the target device.

1. Power up the computer you're installing Windows 10 on and wait for it to detect the USB drive. If this doesn't happen, reboot, this time pressing the key to access the UEFI/BIOS or boot menu. Ensure the USB device is detected, then select it as the main boot device.

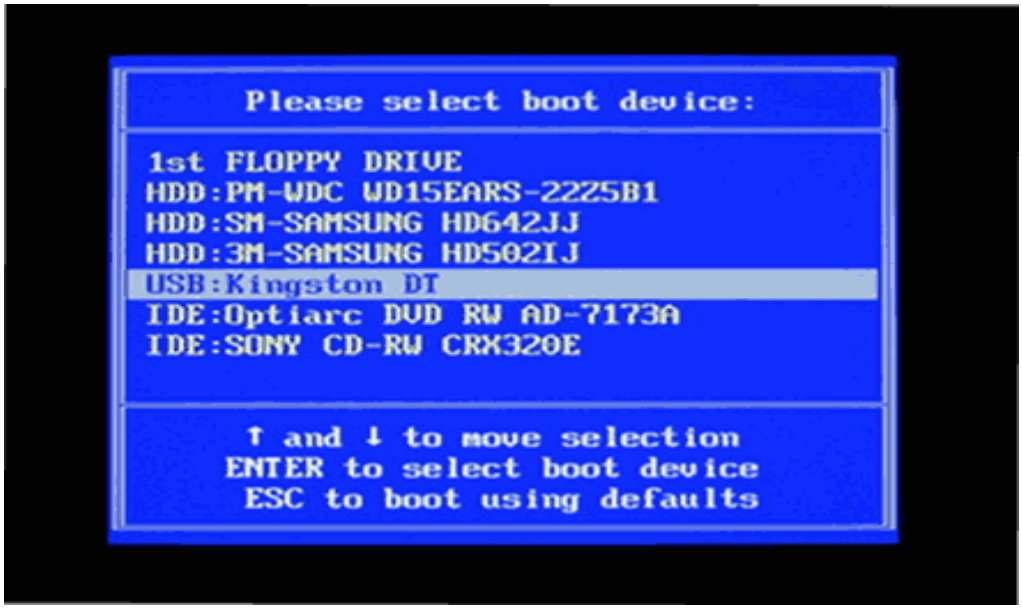


Figure 1.0 ***Selecting bootable device***

2. The subsequent reboot should detect the Windows 10 installation media. You're now ready to install Windows 10! Get started by starting the installation wizard.

Figure 1.1 ***Starting Windows Installer Wizard***

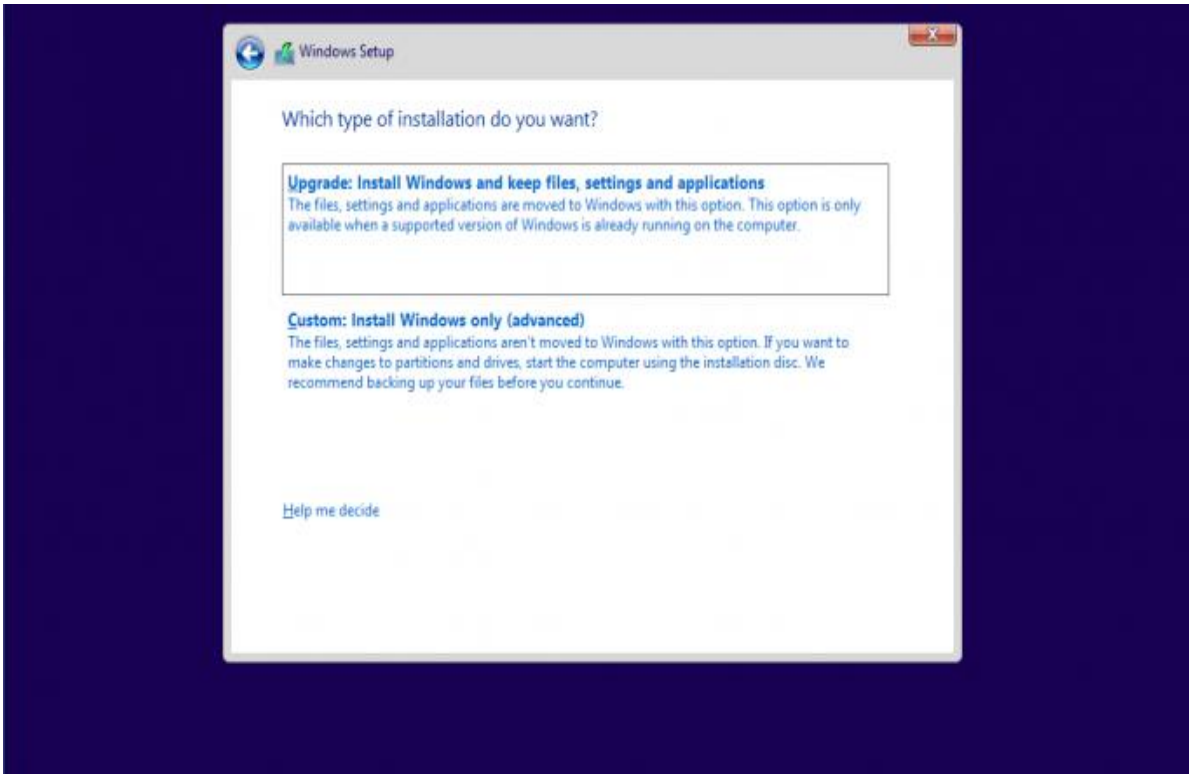


Figure 1.2 ***Selecting the modes of installation***

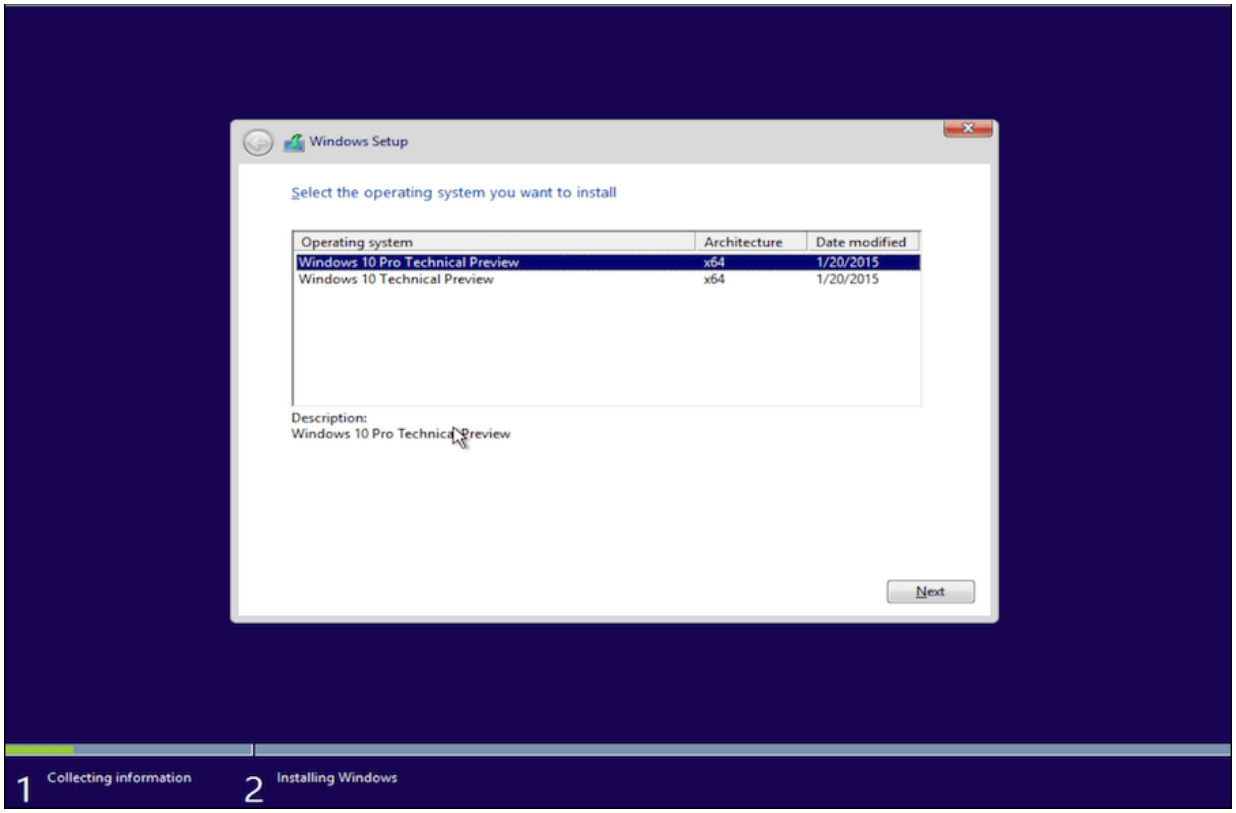


Figure 1.3 ***Selecting Windows Version***

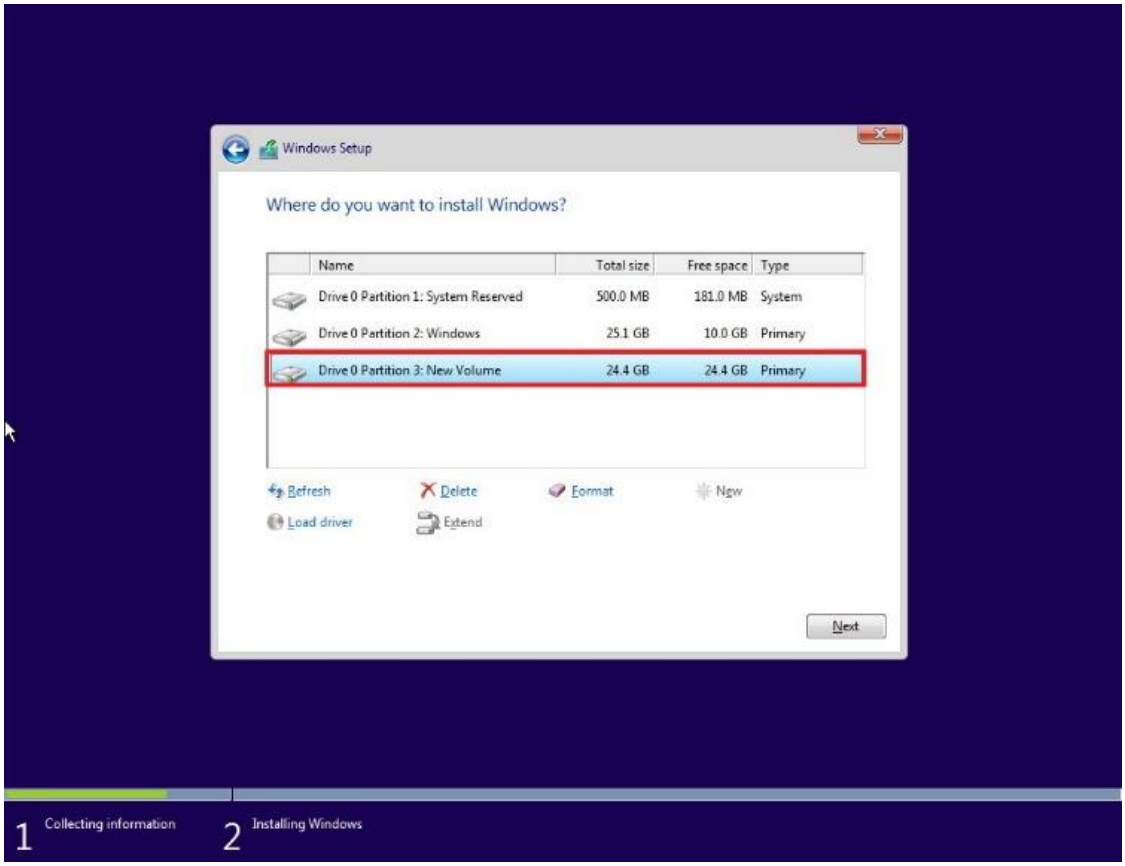


Figure 1.4 ***Partition the storage***

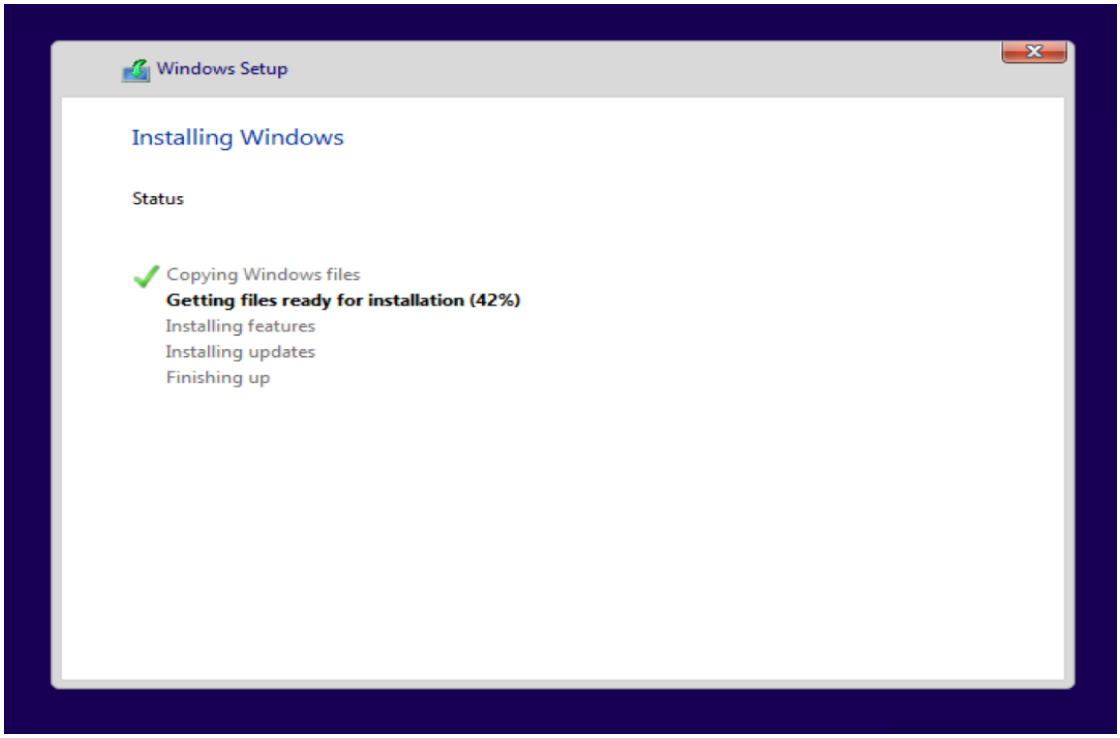


Figure 1.5 ***Starting the installation***

3. Once you've worked through the wizard, Windows 10 will be installed. Note that some installation may continue after you log in, so be patient. It's also worth checking for Windows Updates (Settings > Updates & Security > Windows Update) after installation to ensure everything is up to date.

CREATING ALL IN ONE DRIVER

Downloading all drivers may require ample time and wide variety resource. It may occupy large space on a disc. And also, there are hundreds of different devices which may requires hundreds of different drivers. As a technician we need to have all drivers in our hand which is impossible to obtain. But using Driver Pack application which enable us to possess most recent drivers for different kinds of devices. With this driver we can install any drivers for any devices in any point of time.

Procedure in Preparing DriverPack (All-in-One Driver)

1. Download DriverPack app at https://drp.su/en/foradmin

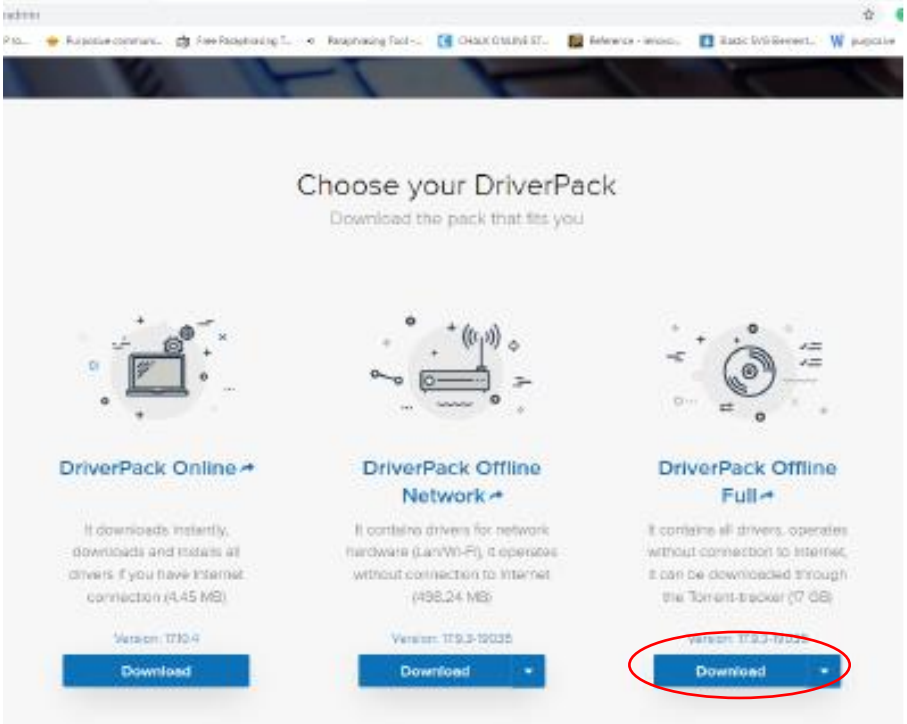


Figure 3.2 ***Screenshot of download page of DriverPack solution website***

2. You will see a "DriverPack-Offline.torrent" downloaded file and the bottom corner of the browser then click to install if download file is not visible press CTRL + J to view downloaded files.

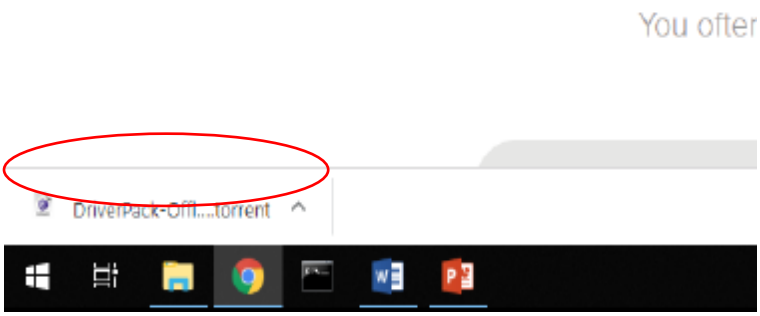


Figure 3.3 ***Screenshot of downloaded torrent file using google chrome***

Note: *You need a torrent downloader such as, bitTorrent/uTorrent in order to download the torrent file. If you don't have installed torrent app you can download bitTorrent application at http://www.bittorrent.com/ after downloading the file you need to install this tiny app.*

Visit http://www.wikihow.com/Install-the-BitTorrent-Client to proceed the installation. **Note: Using bitTorrent downloader is solely use in this procedure to download free application. Downloading cracked/preactivated proprietary software are strongly prohibited**.

3. After clicking the file the image shown below will appear then click ok to download it.

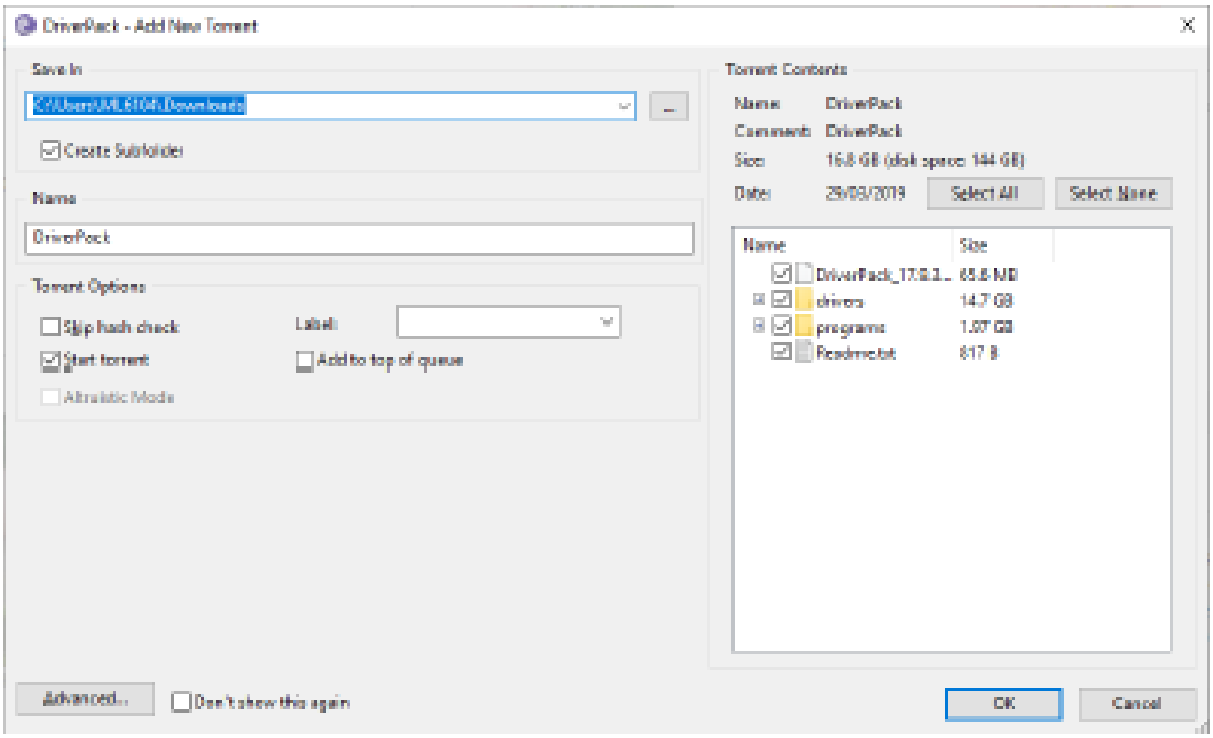


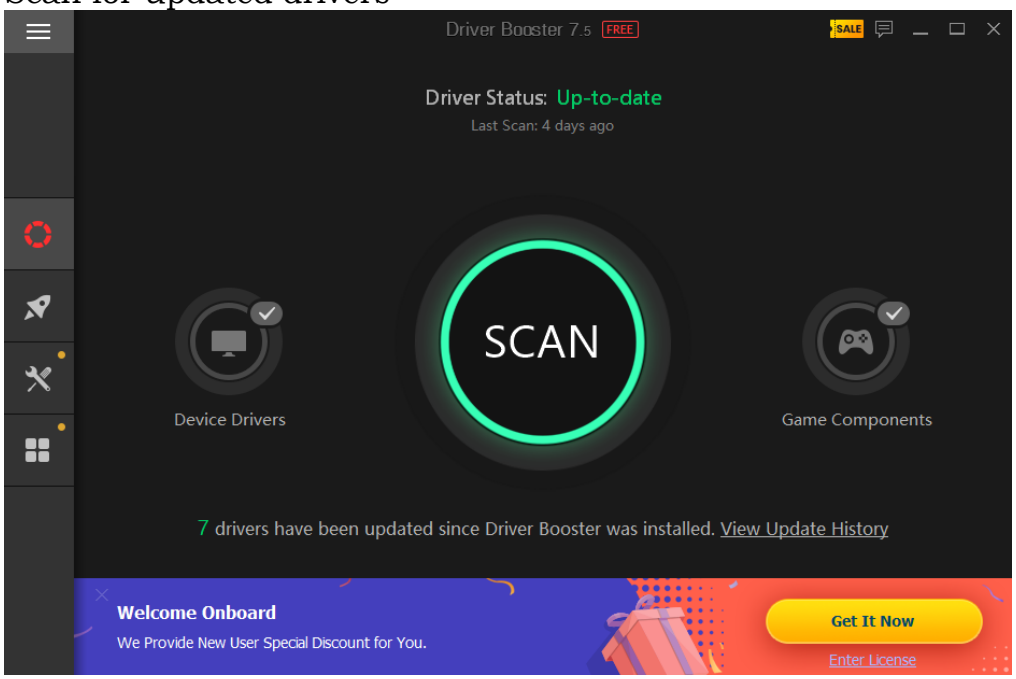
Figure 3.4 ***Dialog box confirm the installation of torrent***

4. The installation is started, wait for several hours to complete the installation.

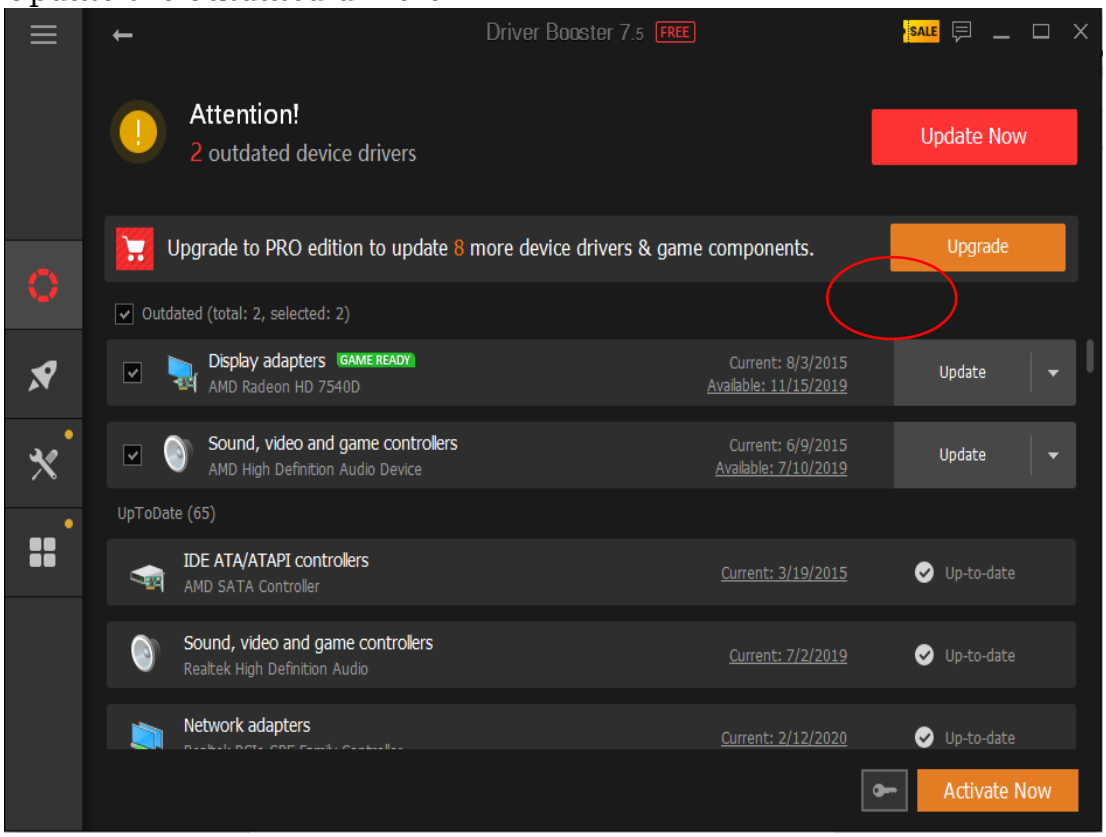
INSTALLING DRIVERS USING UPDATER SOFTWARE (STEP-BY-STEP)

1. Download and install driver booster
   https://www.iobit.com/en/driver-booster.php

2. Scan for updated drivers



3. Update the outdated drivers



What's More

Activity 1.3

I. Encircle the best answer

1. It is a software program that enables a specific hardware device to work with a computer's operating system.

a. Driver b. System software c. application software d. OS

2. It means the devices will work without requiring driver installation.

a. Driver b. OS c. Operating System d. Plug n Play

II. True or False

_____1. We don't need driver during installation/formatting process.
_____2. All devices are not working without installer driver.
_____3. Driver is the linker of the computer's devices attached to it.
_____4. You are trying to reformat a laptop. The first thing you need to do is to obtain its hardware specification.
_____5. Plug and play meaning, the devices will not work without requiring driver installation.
_____6. If system's drivers are not installed, Manufacturers default software are not possible to use
_____7. A device is enabled without a driver.
_____8. Driver is the linker of a certain hardware to computer.
_____9. Scanner is not able to scan without the presence of its required driver.
_____10. Before installation/reformatting we should check first all device drivers needed in installation process.

What I Have Learned

Create a flowchart of the process of preparing a bootable device and installing of Operating System and drivers.

Assessment

Assessment 1.3

Choose the best answer:

1. What os is this

a. windows
b. apple
c. CAPS NEW SHIELDS
d. LINUX

2. What does OS stand for?

a. Optional System
b. Operating system
c. Operand System
d. Operating Structure

3. GUI USES :

a. Commands
b. code
c. menu
d. picture

4. Can computer work without OS?

a. No
b. Yes
c. It will only boot the OS but not load other applications
d. sometimes

5. what is this

a. apples os b. linux c. windows d. ibm

6. A computer uses _____ to organize all of the different files and applications that it contains.

a. Folders/directories
b. Icons
c. Trash Can
d. Applications

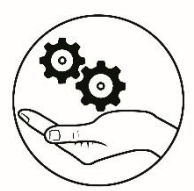
7. What is the purpose of an operating system?

a. Manage software on computer
b. Manage hardware on computer
c. Manage all hardware & software on computer

8. Accessing programs using a Mac OS is different from accessing them on a windows OS in that Mac has a dock instead of a start menu
a. True
b. False

9. A unique, alphanumeric code required by many software programs during installation
a. Product Key
b. Source Code
c. Keying
d. Software Description

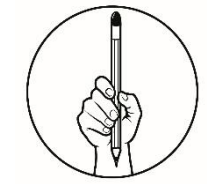
10. Dividing a hard disk drive into multiple logical storage units
a. Partition
b. Division
c. Configure
d. Segment



Additional Activities

- Watch this video for reference:
  https://www.youtube.com/watch?v=SKbR6XT7fcA
  https://www.youtube.com/watch?v=gLfnuE1unS8
- Format your computer using your bootable device and try to install windows 7 and windows 10. Record the process of installing the operating system. Repeat the process until your memorize all the procedures.

Lesson 4 Install application software



What I Know

True or False

1. Desktop App is applications that run in a personal computer and laptops.
2. A web browser is the tool that you use to access the World Wide Web.
3. Application can be put in the computer without using installer.
4. A word processor allows you to write a letter, design a flyer, and create many other kinds of documents. The most well-known word processor is Microsoft Word.
5. If you want to listen to mp3s or watch movies that you've downloaded, you'll need to use a media player.

What's In

On the previous lesson you learn how to install Operating System and driver installer for the computer to work properly. In this lesson, you will learn how to install application that will be useful in using computer and doing our work within the office or personal matters.

What's New

You may have heard people talking about using an application or an app. But what exactly does that mean? An application (or app) is a type of software that allows you to perform specific tasks. Applications for desktop or laptop computers are sometimes called desktop applications, and those for mobile devices are called mobile apps. When you open an application, it runs inside the operating system until you close it. Much of the time, you will have more than one application open at the same time, and this is known as multitasking.

Source: http://www.gcflearnfree.org/computers/computerbasics/

What is It

APPLICATION SOFTWARE

App is a very common term for an application, especially for simple applications that can be downloaded cheaply or even for free. Many apps are also available for mobile devices and even some TVs.

Three types of Application software

A. Desktop App is applications that run in a personal computer and laptops.

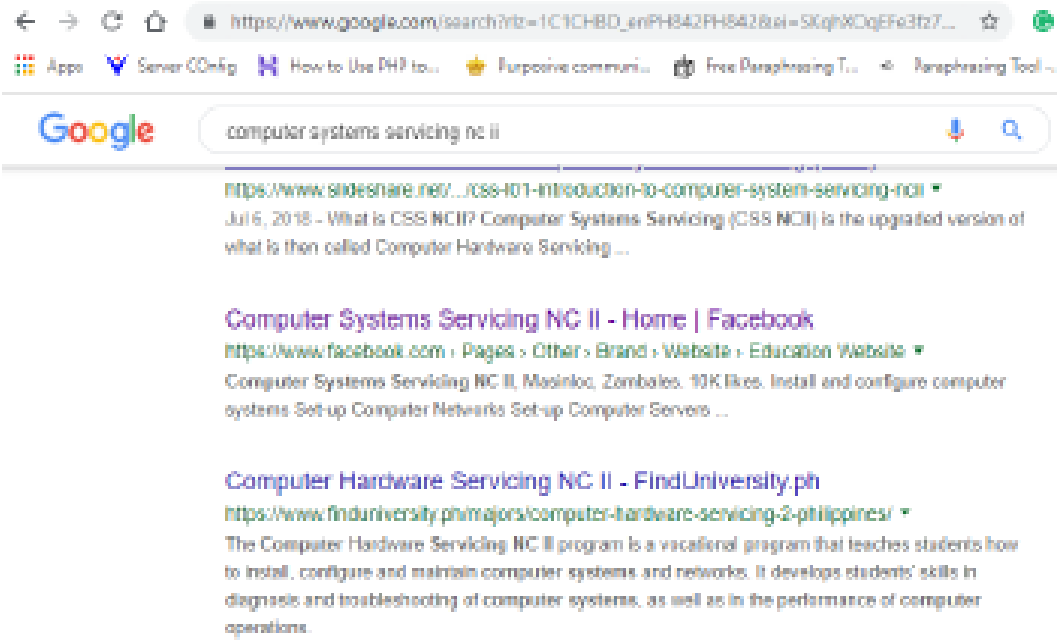
Types of Desktop Applications

There are countless desktop applications out there, and they fall into many different categories. Some are more full-featured (like Microsoft Word), while others may only do one or two things (like gadgets). Below are just a few types of applications that you might use:

- **Word Processors:** A word processor allows you to write a letter, design a flyer, and create many other kinds of documents. The most well-known word processor is Microsoft Word.

- Personal Finance: Personal finance software, such as Quicken, allows you to keep track of your income and expenses, create a budget, and more. Most personal finance programs can automatically download information from your bank, so you don't have to manually type in all of your transactions.

- **Web Browsers:** A web browser is the tool that you use to access the World Wide Web. Most computers come with a web browser pre-installed, but you can also download a different one if you prefer. Examples of browsers include **Internet Explorer, Firefox, Google Chrome, and Safari.**



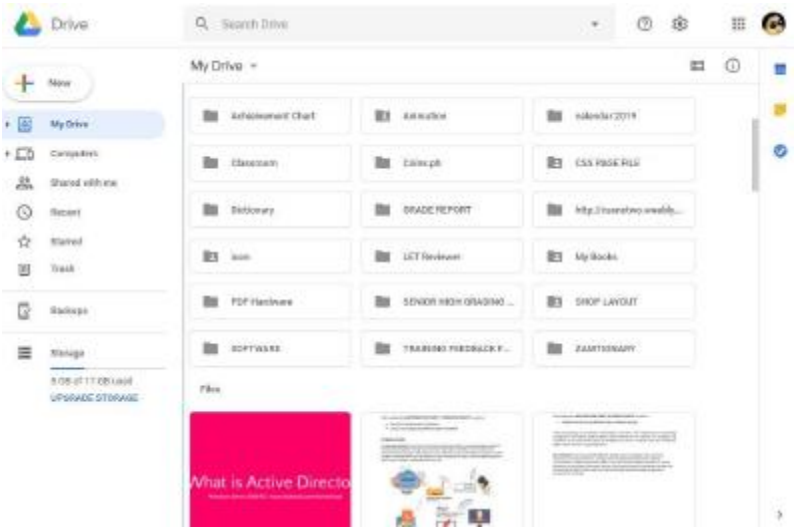
- **Games:** There are many different games that you can play on your computer. They range from card games such as Solitaire, to action games like **PlayerUnkown's Battleground (PUBG)**. Many action games require a lot of computing power, so they may not work unless you have a newer computer.

- **Media Players**: If you want to listen to mp3s or watch movies that you've downloaded, you'll need to use a media player. Netflix and Spotify are popular media services application.

- **Gadgets:** Sometimes called widgets, these are simple applications that you can place on your desktop (or on the Dashboard if you're using a Mac). There are many different types of gadgets, and they include calendars, calculators, maps, news headlines, and more.

B. Web App is an application that is accessed over a network such as the Internet or an intranet.

It run in the cloud and does not need to be installed on your computer. These are sometimes called cloud apps.



Sample of web app service, my google drive screenshot

- **Online Email Services**: Services like Gmail and Yahoo! Mail run within your browser and can do many of the same things that email programs like Microsoft Outlook can do. After you sign up for an online email service, you can begin using it immediately - no installation is required. Instead of being stored on your computer, your emails are stored in the cloud.

- **Google Docs**: Google Docs is an office suite that runs within your browser. Much like Microsoft Office, you can use it to create documents, spreadsheets, presentations, and more. Your documents are stored in the cloud, which makes it easy to share your documents with other people.

- **Facebook**: Facebook lets you create an online profile and interact with your friends. Profiles and conversations are constantly evolving, so Facebook uses web app technologies throughout the site to keep the information up-to-date. There are also games and other web apps that you can add to your Facebook profile.

C. Mobile App is a software application designed to run on smartphones, tablet computers and other mobile devices.

Desktop and laptop computers aren't the only devices that can run apps. You can also download apps for mobile devices like smartphones and tablet computers, which opens up a lot of new possibilities. Here are a few examples of mobile apps:

- **Snapchat** lets you easily talk with friends, view Live Stories from around the world, and explore news in Discover.

- **Messenger** instantly reach the people in your life—for free. Messenger is just like texting, but you don't have to pay for every message (it works with your data plan).
  Not just for Facebook friends: Message people in your phone book and just enter a phone number to add a new contact.

- **Instagram** is a simple way to capture and share the world's moments. Follow your friends and family to see what they're up to, and discover accounts from all over the world that are sharing things you love.

PREPARING INSTALLER APPS

Before doing installation, there are some essential factors that we should always know.

4.1 Platform Environment

We don't just install application without knowing its environment. Identifying the platform is very essential. We must to know its OS, chipsets and other hardware component.

Example

You are about to install an office 2019 app with x64 architecture in a computer running with x86 processor. Technically it is incompatible due to dissimilar architecture of your device and application.

4.2 Categories of Installer Apps

a) Open Source/Freeware are applications that can be use by the end users for free. These applications don't have equivalent price.

b) Closed Source/Paid Apps /Proprietary software

is computer software licensed under exclusive legal right of the copyright holder with the intent that the licensee is given the right to use the software only under certain conditions, and restricted from other uses, such as modification, sharing, studying, redistribution, or reverse engineering.

Are applications that need to purchase before using it such as, Adobe Photoshop, Windows 8 and Microsoft Office.

c) Trial ware

It is also a type of proprietary software which is provided to users for a certain limited trial basis and pursuant to a license which restricts any commercial benefit, use or exploitation of the software. This application usually offers 30 days trial or more.

d) Subscription-based application

A subscription-based is an apps that allows a customer or organization to purchase or subscribe for a specific period of time for a set price. Subscribers typically commit to the services on a monthly or annual basis

Purpose of Installation

Before the installation we need to identify first the objective its execution.

You need a certain apps for a certain purposes. You must to identify if you are really need that particular apps or not.

Example:

You are graphics designer for a clothing product, and you installed Adobe Illustrator because you need to create a vector art of a certain clothing line for your new project. Creating a vector art is the purpose why you installed Adobe Illustrator.

Different Ways of Preparing Installer Apps

There are several ways in preparing installer apps

1. Direct downloading/installation from web

Installation directly from the web is too risky it may contains malware or virus it may also requires ample time in installation mainly if you have poor internet connection.

2. via USB storage devices(e.g., flash disks, external hdd)

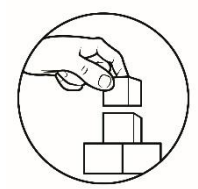
This installation is most recommended compares to other. It is more portable and manageable due to portability of its execution.

3. via installation disc

It is the traditional way of installation of application, it requires ODD in every installation. Software are burned in an optical disc.

4. via intranet or LAN network

This method requires LAN connection in conducting installation. Via intranet installation make faster and more efficient. It enables you to deploy multiple installation during the process.



What's More

Activity 1.4

Direction: Encircle the best answer.

1. It is a type of software that allows you to perform specific tasks.

a. Driver
b. System software
c. Application software
d. Desktop App

2. It is an application that is accessed over a network such as the Internet or an intranet.

a. Web App
b. Mobile App
c. Desktop App
d. System software

3. It is a software application designed to run on smartphones, tablet computers and other mobile devices.

a. Web App
b. Desktop App
c. Mobile App
d. android

4. Free software is also called _____.

a. Closed source
b. Open Source
c. Application software
d. windows

5. Closed software is called _____.

a. Proprietary software
b. Open Source
c. Application software
d. Windows 8

6. Allows you to write a letter, design a flyer, and create many other kinds of documents.

a. Web Browsers
b. Word Processors
c. Personal Finance
d. Media Players

7. Allows you to keep track of your income and expenses, create a budget, and more

a. Personal Finance
b. Media Players
c. Web Browsers
d. Word Processors

8. Is applications that run in a personal computer and laptops.

a. Application software
b. windows
c. Desktop App
d. Web App

9. Is an office suite that runs within your browser.

a. Web Browsers
b. Desktop App
c. Mobile App
d. Google Docs

10. Are applications that run in different platforms using Windows 10 operating system

a. Windows Universal apps
b. Metro Apps/Windows 8.1 Apps
c. Mobile Apps
d. Web Apps

What I Have Learned

What do you think is the importance of installing application on your computer? In what aspect does user benefits in using applications?

Assessment

Assessment 1.4

1. Rose opens her software application and gets a message stating that she has only 16 more uses of the product available. Given this scenario, what is her best course of action in order to use it more than 16 additional times?
    a. Activate the software
    b. Register the software
    c. Uninstall and reinstall the software
    d. Not close the application any more
2. You have just purchased licensed software and want to install it. Which of the following are you required to enter to install this software?

a. Your name and contact information
b. The product key
c. The serial number of your processor
d. None of the above

3. John purchases a single-user software license for an application and installs it on his desktop. Then he installs it on it his laptop computer. In this scenario, which of the following statements is true?
    a. John can use the application on both computers at the same time
    b. John can use the application on both computers but only one at a time
    c. John can use the application on only one of the computers
    d. John can share the application with his friends to install on their computers
4. Word processing, spreadsheets, and presentation software are examples of what?
    a. Productivity software
    b. Collaboration software
    c. Utility software
    d. Operating system add-ins
5. When thinking of software compatibility, which two factors matter most?
    a. If the software will work with other software on the computer
    b. if the software will work with the operating systems
    c. if the minimum hardware requirements are met
    d. If the software comes with automatic updates
6. Your boss calls you on the phone while he is trying to install software. It is telling him that he needs a product key. Where should you tell him to look for it?
    a. On the package the installation cd came in
    b. ON the screen in the options menu
    c. In the windows updates
    d. On the internet
7. Michael receives an email from an unknown user with a file attachment in it that has an .exe extension. What should Michael do with this file?
    a. delete the email
    b. double-click the file to open
    c. save the file to his desktop and then open it.
    d. right click the file to identify its product key and install it
8. You have just plugged in a new HP printer to your Mac. You can't find a printer driver. What should you do given the circumstances?
    a. Uses the default HP printer driver for Mac OS X
    b. Visit Apple's website to download the printer driver
    c. Visit HP's website to download the printer driver
    d. Run Software Update to automatically install the printer driver
9. Which of the following file extensions is a worksheets created in a Microsoft spreadsheet application likely to have?
    a. .spd
    b. .ppt
    c. .mss
    d. .xls
10. You just installed an antivirus program on your laptop computer. Given a scenario in which you want to maintain proper levels of security, how often should you update the software?
    a. At least once a week
    b. At least once a month
    c. At least twice a year
    d. Only when a new virus is discovered

Additional Activities

- Install the essential application that you will going to use in computer. Identify the function and how does application helps the user to do its task.

Answer Key

Pre-Assessment 1.4	Activity 1.4	Assessment 1.4
1. TRUE 2. TRUE 3. FALSE 4. TRUE 5. TRUE	1. A 2. A 3. C 4. B 5. A 6. B 7. A 8. C 9. D 10. A	1. B 2. B 3. C 4. A 5. B & C 6. A 7. A 8. C 9. D 10. A

Assessment 1.2

Multiple Choice

1. A
2. C
3. C
4. A
5. A

Enumeration

1-3. iOs8, Windows Phone 8.1,10, Android Nougat

4-5. Windows 7, Windows 8.1,10, Mac OS X Yosemite, El Capitan Linux Ubuntu, Windows Xp

Pre-Assessment 1.3

1. TRUE
2. FALSE
3. TRUE
4. TRUE
5. TRUE

Activity 1.3

(Multiple Choice)
1. A
2. C

(True or False)
1. FALSE
2. FALSE
3. TRUE
4. TRUE
5. FALSE
6. TRUE
7. FALSE
8. TRUE
9. TRUE
10. TRUE

Assessment 1.3

1. A
2. B
3. D
4. A
5. A
6. A
7. C
8. A
9. A
10. A

Pre-Assessment 1.1

1. B
2. C
3. D
4. A
5. C
6. A
7. D
8. B
9. A
10. D

Activity 1.1

1. MONITOR
2. SPEAKER
3. WEBCAMERA
4. HEADSET
5. MICROPHONE
6. MOUSE
7. FLASHDRIVE
8. POWERCORD
9. KEYBOARD

Assessment 1.1

1. TRUE
2. FALSE
3. TRUE
4. TRUE
5. TRUE
6. FALSE
7. FALSE
8. TRUE
9. TRUE
10. TRUE

Pre-Assessment 1.2

1. TRUE
2. FALSE
3. FALSE
4. FALSE
5. TRUE

Activity 1.2

1. A
2. A
3. B
4. C
5. A
6. A
7. D
8. C
9. B
10. A

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